1. Largest valid value of r:

   $r = 4$

   Example of win records:

   81,76,71,66,61,56,51,46,40,35,30,25,20,15,10,5

   Justification:

   The teams must be ranked s.t. $W_1 > W_2 > W_3 > ... > W_{16}$, where W_i is the number of wins of team i.
   16 total teams means there are 15 gaps between the teams.
   For no team to care about improving their rank, the gap between each team must be at least r+1 which leads to: $W_1 - W_{16} \geq 15(r + 1)$
   After 90 - r rounds, the largest possible difference between the first and last team is: $90 - r \geq 15(r + 1)$
   Solving this inequality for r gives $r \leq 4.6875$.
   And because r must be an integer, $r = 4$ is the largest valid value of r.

   The pigeonhole principle is shows why r cannot be made larger than 4.
   If r were 5, with 15 gaps between the teams, there would need to be a total spread of 90: $15 \cdot 6 = 90$.
   But with only 5 rounds remaining, the maximum spread is 85.
   Thus, if we take the possible win totals of the teams as the holes and the teams as the pigeons, then, with 5 rounds remaining, there are only 86 possible win totals (0 through 85).
   For no team to care about improving their rank, team wins must differ by at least 6, which means that there can only be 15 possible win totals (0,6,12,...,90).
   Thus, there are not enough holes to place all 16 teams while maintaing the required spacing of wins.

2. Simplify the following expression, making one modification per line.
   On each line you should also briefly explain what rule was used to get that line from the one above.
   $\neg(\neg(p \wedge q) \vee \neg(\neg q \wedge r))$

   1. $\neg((\neg p \vee \neg q) \vee \neg(\neg q \wedge r)) \rightarrow$ Demorgan's on $\neg(p \wedge q)$
   2. $\neg((\neg p \vee \neg q) \vee (q \vee \neg r)) \rightarrow$ Demorgan's on $\neg(\neg q \wedge r)$
   3. $\neg(\neg p \vee \neg q \vee q \vee \neg r) \rightarrow$ Associative Law
   4. $p \wedge q \wedge \neg q \wedge r \rightarrow$ Demorgan's on outer parenthesis
   5. $p \wedge F \wedge r \rightarrow q \wedge \neg q$ annihilates to F
   6. $F \rightarrow p \wedge F \wedge r$ is always F

3. Definition: $aXb = \neg(a \vee b)$

Show that we can simulate the following three expressions, with other expressions in which the only operator used is X. The expressions shouldn't include T, F or negation either. You should provide step-by-step derivation with comments. Each change should be on a new line. Each line should be equivalent to the preceding line.

a) $\neg p$

$\neg p \leftrightarrow \neg p \vee \neg p$ as $p \vee p$ is always p

$\neg p \vee \neg p \leftrightarrow \neg(p \vee p)$ by deMorgan's laws

$\neg(p \vee p) \leftrightarrow pXp$ by definition

b) $p \vee q$

$p \vee q \iff \neg(\neg(p \vee q))$ a double logical not cancels out via deMorgan's laws
$\iff \neg(pXq)$ the X operator is defined with a logical not
$\iff (pXq)X(pXq)$ as $pXp \iff \neg p$ as proved above

c) $p \wedge q$

$p \wedge q \leftrightarrow \neg((\neg(p \vee p)) \vee (\neg(q \vee q))$ assumed for the sake of argument
$(\neg p)X(\neg q) \leftrightarrow \neg(\neg p \vee \neg q)$ as $pXp \leftrightarrow \neg p$, which was proven above
$\neg(\neg p \vee \neg q) \leftrightarrow p \wedge q$ via truth table as:

p	q	$\neg(\neg p \vee \neg q)$
T	T	T
T	F	F
F	T	F
F	F	F

$\leftrightarrow$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F